

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

Product Description: Aluminum rod, alloy 6061  
Cat No. : 42053

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

### 1.3. Details of the supplier of the safety data sheet

#### Company

Thermo Fisher (Kandel) GmbH  
Erlenbachweg 2, 76870 Kandel, Germany  
Tel: +49 (0) 721 84007 280  
Fax: +49 (0) 721 84007 300

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

E-mail address begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

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Based on available data, the classification criteria are not met

## Health hazards

Based on available data, the classification criteria are not met

## Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

## 2.2. Label elements

None required

EUH210 - Safety data sheet available on request

## 2.3. Other hazards

In accordance with Annex XIII of the REACH Regulation, inorganic substances do not require assessment

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.2. Mixtures

| Component | CAS No    | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                   |
|-----------|-----------|-------------------|----------|---------------------------------------------------------------------|
| Aluminum  | 7429-90-5 | EEC No. 231-072-3 | 97.9     | -                                                                   |
| Magnesium | 7439-95-4 | EEC No. 231-104-6 | 1.0      | Flam. Sol. 1 (H228)<br>Water-react. 2 (H261)<br>Self-heat. 2 (H252) |
| Silicon   | 7440-21-3 | EEC No. 231-130-8 | 0.6      | -                                                                   |
| Copper    | 7440-50-8 | EEC No. 231-159-6 | 0.27     | -                                                                   |
| Chromium  | 7440-47-3 | EEC No. 231-157-5 | 0.2      | -                                                                   |

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|              |                                                                                                                      |
|--------------|----------------------------------------------------------------------------------------------------------------------|
| Eye Contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician. |
| Skin Contact | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes.                |
| Ingestion    | Clean mouth with water and drink afterwards plenty of water.                                                         |
| Inhalation   | Remove to fresh air.                                                                                                 |

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**Self-Protection of the First Aider** Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

## **4.2. Most important symptoms and effects, both acute and delayed**

No information available.

## **4.3. Indication of any immediate medical attention and special treatment needed**

**Notes to Physician** Treat symptomatically.

## **SECTION 5: FIREFIGHTING MEASURES**

### **5.1. Extinguishing media**

**Suitable Extinguishing Media**  
approved class D extinguishers.

**Extinguishing media which must not be used for safety reasons**  
Water may be ineffective.

### **5.2. Special hazards arising from the substance or mixture**

Thermal decomposition can lead to release of irritating gases and vapors.

**Hazardous Combustion Products**  
None under normal use conditions.

### **5.3. Advice for firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## **SECTION 6: ACCIDENTAL RELEASE MEASURES**

### **6.1. Personal precautions, protective equipment and emergency procedures**

No special precautions required.

### **6.2. Environmental precautions**

Do not flush into surface water or sanitary sewer system.

### **6.3. Methods and material for containment and cleaning up**

Pick up and transfer to properly labelled containers.

### **6.4. Reference to other sections**

Refer to protective measures listed in Sections 8 and 13.

## **SECTION 7: HANDLING AND STORAGE**

### **7.1. Precautions for safe handling**

Ensure adequate ventilation.

**Hygiene Measures**

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Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 7.2. Conditions for safe storage, including any incompatibilities

Keep in a dry place. Keep away from acids.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Storage Class/LGK 13

**Switzerland - Storage of hazardous substances**

Storage class - SC 11/13

<https://www.kvu.ch/de/themen/stoffe-und-produkte>

<https://www.kvu.ch/fr/themes/substances-et-produits>

<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

## 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component | European Union                 | The United Kingdom                                                                                                                         | France                                                                                                                                 | Belgium                                                              | Spain                                             |
|-----------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------|
| Aluminum  |                                | STEL: 30 mg/m <sup>3</sup> 15 min<br>STEL: 12 mg/m <sup>3</sup> 15 min<br>TWA: 10 mg/m <sup>3</sup> 8 hr<br>TWA: 4 mg/m <sup>3</sup> 8 hr  | TWA / VME: 10 mg/m <sup>3</sup><br>(8 heures). metal<br>TWA / VME: 5 mg/m <sup>3</sup> (8<br>heures).                                  | TWA: 1 mg/m <sup>3</sup> 8 uren                                      | TWA / VLA-ED: 1 mg/m <sup>3</sup><br>(8 horas)    |
| Silicon   |                                | STEL: 30 ppm 15 min<br>STEL: 12 mg/m <sup>3</sup> 15 min<br>TWA: 10 mg/m <sup>3</sup> 8 hr<br>TWA: 4 mg/m <sup>3</sup> 8 hr                | TWA / VME: 10 mg/m <sup>3</sup><br>(8 heures).                                                                                         | TWA: 10 mg/m <sup>3</sup> 8 uren                                     |                                                   |
| Copper    |                                | STEL: 0.6 mg/m <sup>3</sup> 15 min<br>STEL: 2 mg/m <sup>3</sup> 15 min<br>TWA: 1 mg/m <sup>3</sup> 8 hr<br>TWA: 0.2 mg/m <sup>3</sup> 8 hr | TWA / VME: 0.2 mg/m <sup>3</sup><br>(8 heures).<br>TWA / VME: 1 mg/m <sup>3</sup> (8<br>heures).<br>STEL / VLCT: 2 mg/m <sup>3</sup> . | TWA: 0.2 mg/m <sup>3</sup> 8 uren<br>TWA: 1 mg/m <sup>3</sup> 8 uren | TWA / VLA-ED: 0.01<br>mg/m <sup>3</sup> (8 horas) |
| Chromium  | TWA: 2 mg/m <sup>3</sup> (8hr) | STEL: 1.5 mg/m <sup>3</sup> 15 min<br>TWA: 0.5 mg/m <sup>3</sup> 8 hr                                                                      | TWA / VME: 2 mg/m <sup>3</sup> (8<br>heures). indicative limit                                                                         | TWA: 0.5 mg/m <sup>3</sup> 8 uren                                    | TWA / VLA-ED: 2 mg/m <sup>3</sup><br>(8 horas)    |

| Component | Italy                                                      | Germany                                                                                                                                                                                                                                          | Portugal                                                               | The Netherlands                   | Finland                                   |
|-----------|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------|-------------------------------------------|
| Aluminum  |                                                            | TWA: 1.25 mg/m <sup>3</sup> (8<br>Stunden). AGW -<br>exposure factor 2<br>TWA: 10 mg/m <sup>3</sup> (8<br>Stunden). AGW -<br>exposure factor 2<br>TWA: 4 mg/m <sup>3</sup> (8<br>Stunden). MAK<br>TWA: 1.5 mg/m <sup>3</sup> (8<br>Stunden). MAK | TWA: 1 mg/m <sup>3</sup> 8 horas                                       |                                   |                                           |
| Copper    |                                                            | TWA: 0.01 mg/m <sup>3</sup> (8<br>Stunden). MAK<br>Höhepunkt: 0.02 mg/m <sup>3</sup>                                                                                                                                                             | TWA: 0.2 mg/m <sup>3</sup> 8 horas<br>TWA: 1 mg/m <sup>3</sup> 8 horas | TWA: 0.1 mg/m <sup>3</sup> 8 uren | TWA: 0.02 mg/m <sup>3</sup> 8<br>tunteina |
| Chromium  | TWA: 0.5 mg/m <sup>3</sup> 8 ore.<br>Time Weighted Average | TWA: 2 mg/m <sup>3</sup> (8<br>Stunden). AGW -<br>exposure factor 1                                                                                                                                                                              | TWA: 0.5 mg/m <sup>3</sup> 8 horas                                     | TWA: 0.5 mg/m <sup>3</sup> 8 uren | TWA: 0.5 mg/m <sup>3</sup> 8<br>tunteina  |

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| Component | Austria                                                                                                                                                                      | Denmark                                                                                                                                                      | Switzerland                                                                    | Poland                                                                           | Norway                                                                                                                                                                                                   |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aluminum  | MAK-KZGW: 20 mg/m <sup>3</sup> 15 Minuten<br>MAK-TMW: 10 mg/m <sup>3</sup> 8 Stunden                                                                                         | TWA: 5 mg/m <sup>3</sup> 8 timer<br>TWA: 2 mg/m <sup>3</sup> 8 timer<br>STEL: 10 mg/m <sup>3</sup> 15 minutter<br>STEL: 4 mg/m <sup>3</sup> 15 minutter      | TWA: 3 mg/m <sup>3</sup> 8 Stunden<br>TWA: 10 mg/m <sup>3</sup> 8 Stunden      | TWA: 2.5 mg/m <sup>3</sup> 8 godzinach<br>TWA: 1.2 mg/m <sup>3</sup> 8 godzinach | TWA: 5 mg/m <sup>3</sup> 8 timer<br>STEL: 10 mg/m <sup>3</sup> 15 minutter.<br>pyrotechnical;value calculated powder                                                                                     |
| Silicon   |                                                                                                                                                                              | TWA: 10 mg/m <sup>3</sup> 8 timer<br>STEL: 20 mg/m <sup>3</sup> 15 minutter                                                                                  | TWA: 3 mg/m <sup>3</sup> 8 Stunden                                             |                                                                                  | TWA: 10 mg/m <sup>3</sup> 8 timer<br>STEL: 20 mg/m <sup>3</sup> 15 minutter. set equal to the limit value for Nuisance dust;value calculated                                                             |
| Copper    | MAK-KZGW: 4 mg/m <sup>3</sup> 15 Minuten<br>MAK-KZGW: 0.4 mg/m <sup>3</sup> 15 Minuten<br>MAK-TMW: 1 mg/m <sup>3</sup> 8 Stunden<br>MAK-TMW: 0.1 mg/m <sup>3</sup> 8 Stunden | TWA: 1.0 mg/m <sup>3</sup> 8 timer<br>TWA: 0.1 mg/m <sup>3</sup> 8 timer<br>STEL: 2 mg/m <sup>3</sup> 15 minutter<br>STEL: 0.2 mg/m <sup>3</sup> 15 minutter | STEL: 0.2 mg/m <sup>3</sup> 15 Minuten<br>TWA: 0.1 mg/m <sup>3</sup> 8 Stunden | TWA: 0.2 mg/m <sup>3</sup> 8 godzinach                                           | TWA: 0.1 mg/m <sup>3</sup> 8 timer<br>TWA: 1 mg/m <sup>3</sup> 8 timer<br>STEL: 3 mg/m <sup>3</sup> 15 minutter. value calculated dust<br>STEL: 0.3 mg/m <sup>3</sup> 15 minutter. value calculated fume |
| Chromium  | MAK-TMW: 2 mg/m <sup>3</sup> 8 Stunden                                                                                                                                       | TWA: 0.5 mg/m <sup>3</sup> 8 timer<br>STEL: 1 mg/m <sup>3</sup> 15 minutter                                                                                  | TWA: 0.5 mg/m <sup>3</sup> 8 Stunden                                           | TWA: 0.5 mg/m <sup>3</sup> 8 godzinach                                           | TWA: 0.5 mg/m <sup>3</sup> 8 timer<br>STEL: 1.5 mg/m <sup>3</sup> 15 minutter. value calculated                                                                                                          |

| Component | Bulgaria                                                  | Croatia                                                                                                                                                   | Ireland                                                                                                                                                                             | Cyprus                   | Czech Republic                                                                                                                                                       |
|-----------|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aluminum  | TWA: 10.0 mg/m <sup>3</sup><br>TWA: 1.5 mg/m <sup>3</sup> | TWA-GVI: 10 mg/m <sup>3</sup> 8 satima. total dust, inhalable particles<br>TWA-GVI: 4 mg/m <sup>3</sup> 8 satima. respirable dust                         | TWA: 1 mg/m <sup>3</sup> 8 hr. respirable fraction<br>STEL: 3 mg/m <sup>3</sup> 15 min                                                                                              |                          | TWA: 10.0 mg/m <sup>3</sup> 8 hodinách. dust                                                                                                                         |
| Silicon   |                                                           | TWA-GVI: 10 mg/m <sup>3</sup> 8 satima. total dust, inhalable particles<br>TWA-GVI: 4 mg/m <sup>3</sup> 8 satima. respirable dust                         | TWA: 4 mg/m <sup>3</sup> 8 hr. respirable dust<br>TWA: 10 mg/m <sup>3</sup> 8 hr. Si total inhalable dust<br>STEL: 30 mg/m <sup>3</sup> 15 min<br>STEL: 12 mg/m <sup>3</sup> 15 min |                          |                                                                                                                                                                      |
| Copper    | TWA: 0.1 mg/m <sup>3</sup>                                | TWA-GVI: 0.2 mg/m <sup>3</sup> 8 satima. Cu fume<br>TWA-GVI: 1 mg/m <sup>3</sup> 8 satima. Cu dust<br>STEL-KGVI: 2 mg/m <sup>3</sup> 15 minutama. dust Cu | TWA: 0.2 mg/m <sup>3</sup> 8 hr. Cu fume<br>TWA: 1 mg/m <sup>3</sup> 8 hr. Cu dusts and mists<br>STEL: 2 mg/m <sup>3</sup> 15 min<br>STEL: 0.6 mg/m <sup>3</sup> 15 min             |                          | TWA: 1 mg/m <sup>3</sup> 8 hodinách. dust<br>TWA: 0.1 mg/m <sup>3</sup> 8 hodinách. fume<br>Ceiling: 2 mg/m <sup>3</sup> dust<br>Ceiling: 0.2 mg/m <sup>3</sup> fume |
| Chromium  | TWA: 2.0 mg/m <sup>3</sup>                                | TWA-GVI: 2 mg/m <sup>3</sup> 8 satima. Cr                                                                                                                 | TWA: 2 mg/m <sup>3</sup> 8 hr.<br>STEL: 6 mg/m <sup>3</sup> 15 min                                                                                                                  | TWA: 2 mg/m <sup>3</sup> | TWA: 0.5 mg/m <sup>3</sup> 8 hodinách. dust<br>Ceiling: 1.5 mg/m <sup>3</sup>                                                                                        |

| Component | Estonia                                                                                                   | Gibraltar                     | Greece                                                                              | Hungary                                                                                                                             | Iceland                                                                                                                                                                                                                                                        |
|-----------|-----------------------------------------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aluminum  | TWA: 10 mg/m <sup>3</sup> 8 tundides. total dust<br>TWA: 4 mg/m <sup>3</sup> 8 tundides. respirable dust  |                               | TWA: 10 mg/m <sup>3</sup><br>TWA: 5 mg/m <sup>3</sup>                               | TWA: 1 mg/m <sup>3</sup> 8 órában. AK                                                                                               | STEL: 10 mg/m <sup>3</sup> dust and powder<br>TWA: 5 mg/m <sup>3</sup> 8 klukkustundum. dust and powder                                                                                                                                                        |
| Silicon   | TWA: 10 mg/m <sup>3</sup> 8 tundides.<br>TWA: 5 mg/m <sup>3</sup> 8 tundides. respirable dust             |                               | TWA: 10 mg/m <sup>3</sup><br>TWA: 5 mg/m <sup>3</sup>                               |                                                                                                                                     | TWA: 10 mg/m <sup>3</sup> 8 klukkustundum.<br>Ceiling: 20 mg/m <sup>3</sup>                                                                                                                                                                                    |
| Copper    | TWA: 1 mg/m <sup>3</sup> 8 tundides. total dust<br>TWA: 0.2 mg/m <sup>3</sup> 8 tundides. respirable dust |                               | STEL: 2 mg/m <sup>3</sup><br>TWA: 0.2 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup> | STEL: 0.2 mg/m <sup>3</sup> 15 percekbén. CK<br>TWA: 0.1 mg/m <sup>3</sup> 8 órában. AK<br>TWA: 0.01 mg/m <sup>3</sup> 8 órában. AK | TWA: 1.0 mg/m <sup>3</sup> 8 klukkustundum. total dust and powder<br>TWA: 0.1 mg/m <sup>3</sup> 8 klukkustundum. Cu respirable fraction, fume<br>Ceiling: 2 mg/m <sup>3</sup> total dust and powder<br>Ceiling: 0.2 mg/m <sup>3</sup> Cu respirable dust, fume |
| Chromium  | TWA: 2 mg/m <sup>3</sup> 8                                                                                | TWA: 2 mg/m <sup>3</sup> 8 hr | TWA: 1 mg/m <sup>3</sup>                                                            | TWA: 2 mg/m <sup>3</sup> 8                                                                                                          | TWA: 0.5 mg/m <sup>3</sup> 8                                                                                                                                                                                                                                   |

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|  |           |  |  |            |                                                                    |
|--|-----------|--|--|------------|--------------------------------------------------------------------|
|  | tundides. |  |  | órában. AK | klukkustundum.<br>powder<br>Ceiling: 1 mg/m <sup>3</sup><br>powder |
|--|-----------|--|--|------------|--------------------------------------------------------------------|

| Component | Latvia                                                  | Lithuania                                                                                                                              | Luxembourg                         | Malta                    | Romania                                                                                                                                         |
|-----------|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Aluminum  | TWA: 2 mg/m <sup>3</sup>                                | TWA: 5 mg/m <sup>3</sup> inhalable fraction IPRD<br>TWA: 2 mg/m <sup>3</sup> respirable fraction IPRD<br>TWA: 1 mg/m <sup>3</sup> IPRD |                                    |                          | TWA: 3 mg/m <sup>3</sup> 8 ore<br>TWA: 1 mg/m <sup>3</sup> 8 ore<br>STEL: 10 mg/m <sup>3</sup> 15 minute<br>STEL: 3 mg/m <sup>3</sup> 15 minute |
| Copper    | STEL: 1 mg/m <sup>3</sup><br>TWA: 0.5 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup> inhalable fraction IPRD<br>TWA: 0.2 mg/m <sup>3</sup> respirable fraction IPRD                                |                                    |                          | TWA: 0.5 mg/m <sup>3</sup> 8 ore<br>STEL: 0.2 mg/m <sup>3</sup> 15 minute<br>STEL: 1.5 mg/m <sup>3</sup> 15 minute                              |
| Chromium  | TWA: 2 mg/m <sup>3</sup>                                | TWA: 2 mg/m <sup>3</sup> IPRD                                                                                                          | TWA: 2 mg/m <sup>3</sup> 8 Stunden | TWA: 2 mg/m <sup>3</sup> | TWA: 2 mg/m <sup>3</sup> 8 ore                                                                                                                  |

| Component | Russia                                                      | Slovak Republic                                                                               | Slovenia                                                                                                      | Sweden                                                                           | Turkey                          |
|-----------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|---------------------------------|
| Aluminum  | TWA: 2 mg/m <sup>3</sup> 0036<br>MAC: 6 mg/m <sup>3</sup>   | TWA: 4 mg/m <sup>3</sup> inhalable dust<br>TWA: 1.5 mg/m <sup>3</sup> respirable dust         |                                                                                                               | TLV: 5 mg/m <sup>3</sup> 8 timmar. NGV<br>TLV: 2 mg/m <sup>3</sup> 8 timmar. NGV |                                 |
| Copper    | TWA: 0.5 mg/m <sup>3</sup> 1234<br>MAC: 1 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup> inhalable fraction<br>TWA: 0.2 mg/m <sup>3</sup> respirable fraction |                                                                                                               | TLV: 0.01 mg/m <sup>3</sup> 8 timmar. NGV                                        |                                 |
| Chromium  |                                                             |                                                                                               | TWA: 2 mg/m <sup>3</sup> 8 urah inhalable fraction<br>STEL: 2 mg/m <sup>3</sup> 15 minutah inhalable fraction | TLV: 0.5 mg/m <sup>3</sup> 8 timmar. NGV                                         | TWA: 2 mg/m <sup>3</sup> 8 saat |

## Biological limit values

List source(s):

| Component | European Union | United Kingdom | France                                                                                                                                          | Spain | Germany                                                                                                     |
|-----------|----------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------------------|
| Aluminum  |                |                |                                                                                                                                                 |       | Aluminum: 50 µg/g Creatinine urine (for long-term exposures: at the end of the shift after several shifts ) |
| Chromium  |                |                | Total Chromium: 0.01 mg/g creatinine urine augmented during shift<br>Total Chromium: 0.03 mg/g creatinine urine end of shift at end of workweek |       |                                                                                                             |

| Component | Italy | Finland | Denmark | Bulgaria | Romania                                                                                                        |
|-----------|-------|---------|---------|----------|----------------------------------------------------------------------------------------------------------------|
| Aluminum  |       |         |         |          | Aluminum: 200 µg/L urine end of shift                                                                          |
| Chromium  |       |         |         |          | Chromium: 10 µg/g Creatinine urine during working hours<br>Chromium: 30 µg/g Creatinine urine end of work week |

| Component | Gibraltar | Latvia                                                            | Slovak Republic                                 | Luxembourg | Turkey |
|-----------|-----------|-------------------------------------------------------------------|-------------------------------------------------|------------|--------|
| Aluminum  |           |                                                                   | Aluminum: 60 µg/g creatinine urine not critical |            |        |
| Chromium  |           | Chromium: 10 µg/g Creatinine urine end of shift; end of work week |                                                 |            |        |

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## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                    | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|------------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| Copper<br>7440-50-8 ( 0.27 ) |                              | DNEL = 273mg/kg<br>bw/day       |                                | DNEL = 137mg/kg<br>bw/day         |

| Component                     | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|-------------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Chromium<br>7440-47-3 ( 0.2 ) |                                  |                                     | DNEL = 0.5mg/m <sup>3</sup>        |                                       |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                      | Fresh water    | Fresh water sediment             | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)          |
|--------------------------------|----------------|----------------------------------|--------------------|------------------------------------|-----------------------------|
| Aluminum<br>7429-90-5 ( 97.9 ) |                |                                  |                    | PNEC = 20mg/L                      |                             |
| Copper<br>7440-50-8 ( 0.27 )   | PNEC = 7.8µg/L | PNEC = 87mg/kg<br>sediment dw    |                    | PNEC = 230µg/L                     | PNEC = 65mg/kg<br>soil dw   |
| Chromium<br>7440-47-3 ( 0.2 )  | PNEC = 6.5µg/L | PNEC = 205.7mg/kg<br>sediment dw |                    |                                    | PNEC = 21.1mg/kg<br>soil dw |

| Component                    | Marine water   | Marine water sediment          | Marine water Intermittent | Food chain | Air |
|------------------------------|----------------|--------------------------------|---------------------------|------------|-----|
| Copper<br>7440-50-8 ( 0.27 ) | PNEC = 5.2µg/L | PNEC = 676mg/kg<br>sediment dw |                           |            |     |

## 8.2. Exposure controls

### Engineering Measures

None under normal use conditions.

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles) (European standard - EN 166)

#### Hand Protection

No special protective equipment required

| Glove material    | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|-------------------|-----------------------------------|-----------------|-------------|-----------------------|
| Disposable gloves | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |

#### Skin and body protection

Long sleeved clothing.

#### Respiratory Protection

No special protective equipment required.

### Large scale/emergency use

In case of insufficient ventilation, wear suitable respiratory equipment

### Small scale/Laboratory use

No personal respiratory protective equipment normally required  
When RPE is used a face piece Fit Test should be conducted

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**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                         |                          |                                   |
|-----------------------------------------|--------------------------|-----------------------------------|
| Physical State                          | Solid                    |                                   |
| Appearance                              | Silver / Grey            |                                   |
| Odor                                    | Odorless                 |                                   |
| Odor Threshold                          | No data available        |                                   |
| Melting Point/Range                     | 660.4 °C                 |                                   |
| Softening Point                         | No data available        |                                   |
| Boiling Point/Range                     | No data available        |                                   |
| Flammability (liquid)                   | Not applicable           | Solid                             |
| Flammability (solid,gas)                | No information available |                                   |
| Explosion Limits                        | No data available        |                                   |
| Flash Point                             | No data available        | Method - No information available |
| Autoignition Temperature                | No data available        |                                   |
| Decomposition Temperature               | No data available        |                                   |
| pH                                      | No data available        |                                   |
| Viscosity                               | No data available        |                                   |
| Water Solubility                        | Insoluble                |                                   |
| Solubility in other solvents            | No information available |                                   |
| Partition Coefficient (n-octanol/water) |                          |                                   |
| Vapor Pressure                          | No data available        |                                   |
| Density / Specific Gravity              | No data available        |                                   |
| Bulk Density                            | 2.7 g/cm3                |                                   |
| Vapor Density                           | No data available        | (Air = 1.0)                       |
| Particle characteristics                | No data available        |                                   |

### 9.2. Other information

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

Yes

### 10.2. Chemical stability

Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

**Hazardous Polymerization** No information available.  
**Hazardous Reactions** No information available.

### 10.4. Conditions to avoid

Incompatible products. Excess heat.

### 10.5. Incompatible materials

None known.

### 10.6. Hazardous decomposition products

None under normal use conditions.



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## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

|                     |                   |
|---------------------|-------------------|
| (a) acute toxicity; |                   |
| Oral                | No data available |
| Dermal              | No data available |
| Inhalation          | No data available |

#### Toxicology data for the components

| Component | LD50 Oral                 | LD50 Dermal | LC50 Inhalation               |
|-----------|---------------------------|-------------|-------------------------------|
| Aluminum  | -                         | -           | LC50 > 0.888 mg/L ( Rat ) 4 h |
| Magnesium | LD50 = 230 mg/kg ( Rat )  | -           | -                             |
| Silicon   | LD50 = 3160 mg/kg ( Rat ) | -           | -                             |
| Copper    | -                         | -           | LC50 > 5.11 mg/L ( Rat ) 4 h  |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;  
    Respiratory No data available  
    Skin No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available  
There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available  
    Target Organs No information available.

(j) aspiration hazard; No data available

Symptoms / effects, both acute and delayed No information available.

### 11.2. Information on other hazards

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

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## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

#### Ecotoxicity effects

Contains a substance which is: Very toxic to aquatic organisms. The product contains following substances which are hazardous for the environment.

| Component | Freshwater Fish                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Water Flea                                    | Freshwater Algae                                                                                                                                   |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Copper    | LC50: = 1.25 mg/L, 96h static (Lepomis macrochirus)<br>LC50: = 0.3 mg/L, 96h semi-static (Cyprinus carpio)<br>LC50: = 0.8 mg/L, 96h static (Cyprinus carpio)<br>LC50: = 0.112 mg/L, 96h flow-through (Poecilia reticulata)<br>LC50: = 0.052 mg/L, 96h flow-through (Oncorhynchus mykiss)<br>LC50: 0.0068 - 0.0156 mg/L, 96h (Pimephales promelas)<br>LC50: < 0.3 mg/L, 96h static (Pimephales promelas)<br>LC50: = 0.2 mg/L, 96h flow-through (Pimephales promelas) | EC50: = 0.03 mg/L, 48h Static (Daphnia magna) | EC50: 0.031 - 0.054 mg/L, 96h static (Pseudokirchneriella subcapitata)<br>EC50: 0.0426 - 0.0535 mg/L, 72h static (Pseudokirchneriella subcapitata) |

### 12.2. Persistence and degradability

#### Degradability

#### Degradation in sewage treatment plant

No information available

Not relevant for inorganic substances.

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

### 12.3. Bioaccumulative potential

No information available

| Component | log Pow | Bioconcentration factor (BCF) |
|-----------|---------|-------------------------------|
| Chromium  |         | 1.03 - 1.22                   |

### 12.4. Mobility in soil

No information available

### 12.5. Results of PBT and vPvB assessment

In accordance with Annex XIII of the REACH Regulation, inorganic substances do not require assessment.

### 12.6. Endocrine disrupting properties

#### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

### 12.7. Other adverse effects

#### Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

#### Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

#### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point. Empty containers

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retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**European Waste Catalogue (EWC)** According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

**Other Information** Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not flush to sewer.

**Switzerland - Waste Ordinance** Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## SECTION 14: TRANSPORT INFORMATION

**IMDG/IMO** Not regulated

14.1. UN number  
14.2. UN proper shipping name  
14.3. Transport hazard class(es)  
14.4. Packing group

**ADR** Not regulated

14.1. UN number  
14.2. UN proper shipping name  
14.3. Transport hazard class(es)  
14.4. Packing group

**IATA** Not regulated

14.1. UN number  
14.2. UN proper shipping name  
14.3. Transport hazard class(es)  
14.4. Packing group

14.5. Environmental hazards No hazards identified

14.6. Special precautions for user No special precautions required.

14.7. Maritime transport in bulk according to IMO instruments Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### **International Inventories**

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component | CAS No    | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|-----------|-----------|-----------|--------|-----|-------|------|----------|------|------|
| Aluminum  | 7429-90-5 | 231-072-3 | -      | -   | X     | X    | KE-00881 | X    | -    |
| Magnesium | 7439-95-4 | 231-104-6 | -      | -   | X     | X    | KE-22673 | X    | -    |
| Silicon   | 7440-21-3 | 231-130-8 | -      | -   | X     | X    | KE-31029 | X    | -    |
| Copper    | 7440-50-8 | 231-159-6 | -      | -   | X     | X    | KE-08896 | X    | -    |
| Chromium  | 7440-47-3 | 231-157-5 | -      | -   | X     | X    | KE-05970 | X    | -    |

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| Component | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|-----------|-----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Aluminum  | 7429-90-5 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Magnesium | 7439-95-4 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Silicon   | 7440-21-3 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Copper    | 7440-50-8 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Chromium  | 7440-47-3 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Authorisation/Restrictions according to EU REACH

| Component | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Aluminum  | 7429-90-5 | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |
| Magnesium | 7439-95-4 | -                                                                   | -                                                                             | -                                                                                                     |
| Silicon   | 7440-21-3 | -                                                                   | -                                                                             | -                                                                                                     |
| Copper    | 7440-50-8 | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |
| Chromium  | 7440-47-3 | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

## REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Seveso III Directive (2012/18/EC)

| Component | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Aluminum  | 7429-90-5 | Not applicable                                                                            | Not applicable                                                                           |
| Magnesium | 7439-95-4 | Not applicable                                                                            | Not applicable                                                                           |
| Silicon   | 7440-21-3 | Not applicable                                                                            | Not applicable                                                                           |
| Copper    | 7440-50-8 | Not applicable                                                                            | Not applicable                                                                           |
| Chromium  | 7440-47-3 | Not applicable                                                                            | Not applicable                                                                           |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

Water endangering class = non-hazardous to waters (self classification)

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| Component | Germany - Water Classification (AwSV) | Germany - TA-Luft Class                               |
|-----------|---------------------------------------|-------------------------------------------------------|
| Aluminum  | nwg                                   |                                                       |
| Magnesium | nwg                                   |                                                       |
| Silicon   | nwg                                   |                                                       |
| Copper    | WGK2                                  | Class III : 1 mg/m <sup>3</sup> (Massenkonzentration) |
| Chromium  | nwg                                   | Class III : 1 mg/m <sup>3</sup> (Massenkonzentration) |

| Component | France - INRS (Tables of occupational diseases)                                                                       |
|-----------|-----------------------------------------------------------------------------------------------------------------------|
| Aluminum  | Tableaux des maladies professionnelles (TMP) - RG 32<br>Tableaux des maladies professionnelles (TMP) - RG 16,RG 16bis |
| Chromium  | Tableaux des maladies professionnelles (TMP) - RG 10                                                                  |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component                     | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|-------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Copper<br>7440-50-8 ( 0.27 )  | Prohibited and Restricted Substances                                                                           |                                                                                 |                                                                                             |
| Chromium<br>7440-47-3 ( 0.2 ) | Prohibited and Restricted Substances                                                                           |                                                                                 |                                                                                             |

## 15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

#### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

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Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

|                       |                       |
|-----------------------|-----------------------|
| Physical hazards      | On basis of test data |
| Health Hazards        | Calculation method    |
| Environmental hazards | Calculation method    |

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

|                  |                                                    |
|------------------|----------------------------------------------------|
| Prepared By      | Health, Safety and Environmental Department        |
| Revision Date    | 20-Feb-2024                                        |
| Revision Summary | New emergency telephone response service provider. |

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,  
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and  
Preparations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**